Chemical Thermodynamics & Kinetics

Distraction-free, high-retention study notes with line-clean dividers, key-value memory grids, derivations & solved CAT-1 / FAT PYQs

Sections: M1S1 Systems & 0th/1st Law • M1S2 Enthalpy & P-V Work • M1S3 2nd Law, Entropy & Carnot • M1S4 Free Energy & 3rd Law • M1S5 Rate Laws (0, 1, 2 Order) • M1S6 Arrhenius & Catalysis • M1S7–S8 Solved Numericals & PYQs

M1S0 Quick-Reference Constants, Conversions & Syllabus Map (7 sections)

Essential Constants to Memorize Before Solving Any Numerical:

- **Universal Gas Constant (R):** $R = 8.314\ J\ K^{-1}\ mol^{-1}$ = 0.08206 L·atm K^{-1} mol^{-1} = 1.987 cal K^{-1} mol^{-1}
- **Unit Conversions (Crucial Exam Trap!): 1 L·atm = 101.325 J | 1 cal = 4.184 J | 1 L·bar = 100 J**
- **Monoatomic Ideal Gas (He, Ne, Ar):** $C_v = (3/2)R = 12.47$ J/K·mol; $C_p = (5/2)R = 20.79$ J/K·mol; $\gamma = C_p/C_v = 5/3 = 1.67$
- **Diatomic Ideal Gas (N_2, O_2, H_2, CO):** $C_v = (5/2)R = 20.79$ J/K·mol; $C_p = (7/2)R = 29.10$ J/K·mol; $\gamma = 7/5 = 1.40$

M1S1 Thermodynamic Systems, Properties, Zeroth Law & First Law (5 core concepts)

Universe = System + Surroundings, separated by a real or imaginary **boundary** (rigid vs. movable, diathermal vs. adiabatic, permeable vs. impermeable).

1. Three System Types (Mass vs. Energy)	• **Open System:** Exchanges **both matter ($\Delta m \neq 0$) and energy ($q \neq 0, w \neq 0$)** across a permeable, diathermal boundary. *Examples:* Boiling water in an open beaker, living cell, human body, turbine. • **Closed System:** Exchanges **energy only ($q \neq 0$ or $w \neq 0$), NO matter ($\Delta m = 0$)** across an impermeable boundary. *Examples:* Gas sealed in a cylinder with a movable piston, sealed hot-water bottle. • **Isolated System:** Exchanges **neither matter nor energy ($\Delta m = 0, q = 0, w = 0$)** across a rigid, adiabatic wall. *Examples:* Ideal sealed thermos flask, insulated bomb calorimeter, the Universe.
2. Intensive vs. Extensive Properties (Size Independence)	• **Intensive Properties (Independent of mass/amount):** Remain unchanged if the system is cut in half → **Temperature (T), Pressure (P), Density ($\rho = m/V$), Boiling Point, Melting Point, Viscosity, Refractive Index, Molar Heat Capacity (C_m), Electrode Potential (E°)**. • **Extensive Properties (Additive / Depend on mass):** Double when amount doubles → **Mass (m), Volume (V), Length, Weight, Internal Energy (U), Enthalpy (H), Entropy (S), Free Energy (A, G), Total Heat Capacity (C)**. • **Memory Hook: Extensive ÷ Extensive = Intensive** (e.g., Density ρ = Mass / Volume; Molar Volume V_m = V / n).
3. State Functions vs. Path Functions (Exact vs. Inexact)	• **State Functions (P, V, T, U, H, S, A, G):** Depend **only on initial and final equilibrium states** ($\Delta X = X_{final} - X_{initial}$; cyclic integral $\oint dX = 0$; exact differentials *dU, dH, dS, dG*). Going from State A to B via Path I or Path II gives the exact same $\Delta U_I = \Delta U_{II}$. • **Path Functions (Heat q & Work w):** Depend on the **specific path/mechanism** taken ($w_I \neq w_{II}$; inexact differentials *đq, đw*; cyclic integral $\oint đw \neq 0$).
4. Zeroth Law of Thermodynamics (Thermometry Basis)	• **Statement:** If body A is in thermal equilibrium with body C ($T_A = T_C$) and body B is separately in thermal equilibrium with body C ($T_B = T_C$), then A and B are in thermal equilibrium with each other ($T_A = T_B$). • **Application:** Defines empirical temperature and forms the working principle of all **thermometers (Body C)** and industrial thermal control loops.

5. First Law & IUPAC Sign Rules (ΔU = q + w)	• **Statement:** Energy can neither be created nor destroyed ($\Delta U_{universe} = 0$); impossible to build a perpetual motion machine of the first kind (PMM-1). • **Equation:** $\Delta U = q + w$ (Differential: $dU = đq + đw$; if *w* denotes work done *by* the system, $\Delta U = q - w$). • **Sign Rules:** Heat absorbed by system: **q > 0 (+)** | Heat released: **q < 0 (−)** | Compression (work ON system, ΔV < 0): **w > 0 (+)** | Expansion (work BY system, ΔV > 0): **w < 0 (−)**. • **P–V Expansion Work:** $dw = -P_{ex}\,dV \Rightarrow w = -\int P_{ex}\,dV$. Against vacuum ($P_{ex} = 0$, free expansion), **w = 0**!

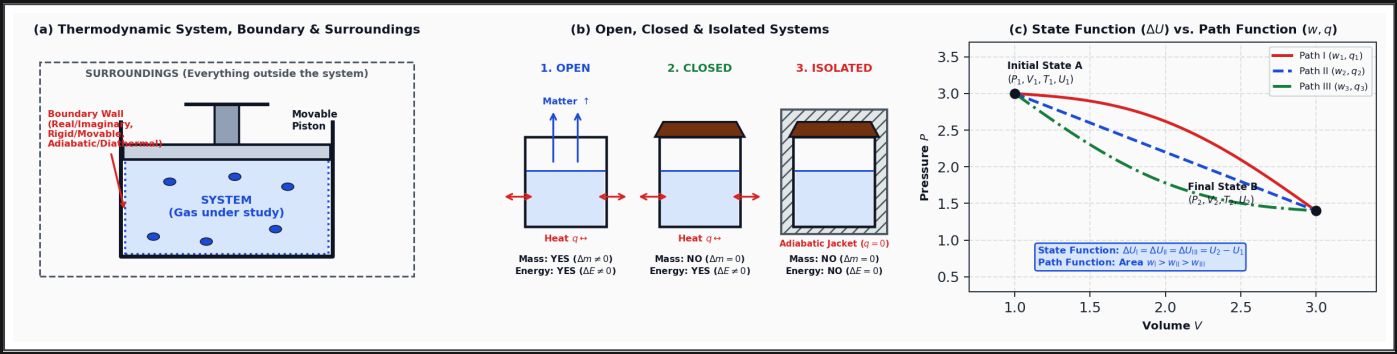


Figure 1.1: (a) System, Boundary & Surroundings; (b) Open, Closed & Isolated systems; (c) State function path independence ($\Delta U_I = \Delta U_{II}$); (d) Zeroth Law thermal equilibrium.

M1S2 Enthalpy, Mayer's Relation & All Four P–V Processes (6 key derivations)

1. Isochoric Heat (q_V) vs. Isobaric Enthalpy (q_P)	• **At Constant Volume (Isochoric, dV = 0):** $w = -P_{ex}(0) = 0 \Rightarrow \Delta U = q_V$ (measured in a rigid bomb calorimeter). • **At Constant Pressure (Isobaric, dP = 0):** Defining Enthalpy $H \equiv U + PV$, we get $\Delta H = q_P = \Delta U + P\Delta V$. • **For Ideal Gas Reactions at Constant T:** Since $P\Delta V = \Delta n_g RT$, $\Delta H = \Delta U + \Delta n_g RT$, where $\Delta n_g = \Sigma n_{g,products} - \Sigma n_{g,reactants}$.
2. Heat Capacities & Derivation of $C_p - C_v = R$ (Frequent 5-Mark Proof)	• **Definitions:** $C_v = (\partial U/\partial T)_V \Rightarrow \Delta U = n C_v \Delta T$ and $C_p = (\partial H/\partial T)_P \Rightarrow \Delta H = n C_p \Delta T$ (valid for an ideal gas in **any** process because *U* and *H* depend on *T* alone!). • **3-Line Proof of Mayer's Relation:** For 1 mole of ideal gas ($PV = RT$), substitute into $H = U + PV$: 1. $H = U + RT$ 2. Differentiate w.r.t. *T*: $(dH/dT) = (dU/dT) + R$ 3. Substitute $C_p = dH/dT$ and $C_v = dU/dT$: $C_p = C_v + R \Rightarrow C_p - C_v = R$.
3. Adiabatic Expansion Derivation ($TV^{\gamma-1} = C$ & Cooling)	• **Reversible Adiabatic (đq = 0):** By 1st Law, $dU = đw \Rightarrow n C_v\,dT = -P\,dV = -(nRT/V)\,dV$. Dividing by *nT* and integrating: $C_v \ln(T_f/T_i) = -R \ln(V_f/V_i) \Leftrightarrow C_v \ln(T_i/T_f) = R \ln(V_f/V_i) \Rightarrow T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$ and $P_i V_i^{\gamma} = P_f V_f^{\gamma}$. • **Why CO_2 Fire Extinguishers Cool Down (Slide 15):** Rapid adiabatic expansion (**q = 0, w < 0**) uses internal energy to do work (**ΔU = w < 0 ⇒ ΔT < 0**), chilling the gas into white frost/dry ice. • **Irreversible Adiabatic Formula (Slide 16):** $C_v(T_f - T_i) = R P_{ex} [(T_i P_f - T_f P_i) / (P_i P_f)]$.

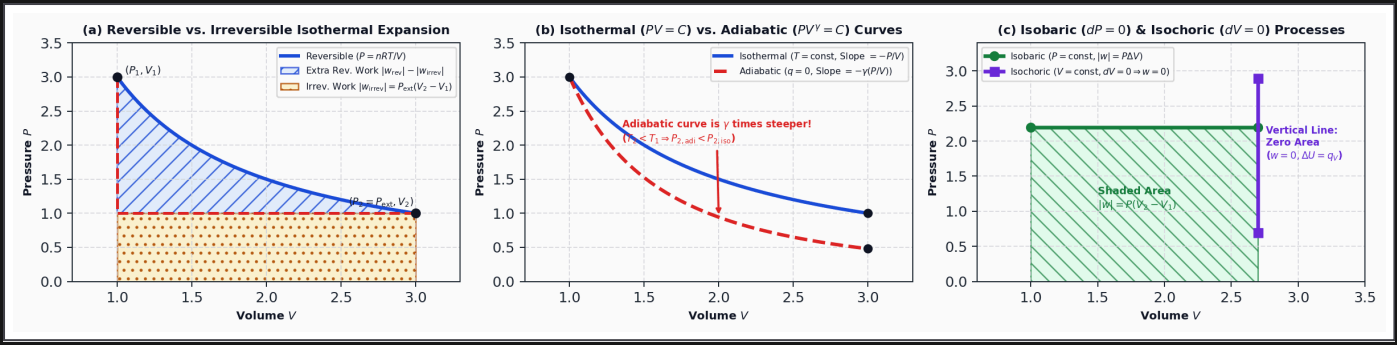


Figure 1.2: (a) Reversible isothermal hyperbola vs. irreversible rectangular work area ($|w_{rev}| > |w_{irrev}|$); (b) Isothermal vs. steeper adiabatic curve ($PV^\gamma = C$); (c) Isobaric rectangle and isochoric vertical line ($w = 0$).

Thermodynamic Process	ΔU & ΔH	Work (w) & Heat (q)	Entropy Change (ΔS_{sys})	Free Energy (ΔG)
Isothermal Reversible ($dT = 0$, $P_{ex} = P$)	$\Delta U = 0$ $\Delta H = 0$	$w_{rev} = -nRT \ln(V_f/V_i) = -nRT \ln(P_i/P_f)$ $q_{rev} = -w_{rev} = +nRT \ln(V_f/V_i)$	$\Delta S = nR \ln(V_f/V_i)$ $= nR \ln(P_i/P_f)$	$\Delta G = nRT \ln(P_f/P_i)$ $= nRT \ln(V_i/V_f)$
Isothermal Irreversible ($dT = 0$, $P_{ex} = const$)	$\Delta U = 0$ $\Delta H = 0$	$w_{irrev} = -P_{ex}(V_f - V_i)$ $q_{irrev} = -w_{irrev} = +P_{ex}(V_f - V_i)$	$\Delta S_{sys} = nR \ln(V_f/V_i)$ (State function!)	$\Delta G = nRT \ln(P_f/P_i)$
Isothermal Free Exp. ($dT = 0$, $P_{ex} = 0$)	$\Delta U = 0, \Delta H = 0$	$w = 0, \ q = 0$	$\Delta S_{sys} = nR \ln(V_f/V_i) > 0$ ($\Delta S_{surr} = 0$)	$\Delta G = -nRT \ln(V_f/V_i)$
Adiabatic Reversible ($đq = 0$, $PV^\gamma = C$)	$\Delta U = nC_v\Delta T$ $\Delta H = nC_p\Delta T$	$q = 0$ $w = \Delta U = nC_v(T_f - T_i) = (P_fV_f - P_iV_i)/(\gamma - 1)$	$\Delta S = 0$ (Isentropic)	Depends on T_i, T_f
Isobaric (dP = 0)	$\Delta U = nC_v\Delta T$ $\Delta H = nC_p\Delta T$	$w = -P(V_f - V_i) = -nR(T_f - T_i)$ $q_P = \Delta H = nC_p(T_f - T_i)$	$\Delta S = nC_p \ln(T_f/T_i)$	Phase tr.: $\Delta G = 0$
Isochoric (dV = 0)	$\Delta U = nC_v\Delta T$ $\Delta H = nC_p\Delta T$	$w = 0$ $q_V = \Delta U = nC_v(T_f - T_i)$	$\Delta S = nC_v \ln(T_f/T_i)$	$\Delta A = \Delta U - T\Delta S$

M1S3 Second Law of Thermodynamics, Entropy & The 4-Stroke Carnot Cycle (4 strokes)

1. Why 1st Law Fails & 2nd Law Statements	• **Limitation of 1st Law:** Conserves total energy but **cannot predict spontaneity/direction** (why heat never flows cold → hot by itself) or why heat cannot be converted 100% into work in a cyclic engine. • **Kelvin–Planck Statement (Heat Engine):** Impossible to build a cyclic engine that absorbs heat from a single reservoir and converts 100% of it into work without rejecting some heat ($|q_c|$) to a colder sink. • **Clausius Statement (Refrigerator):** Impossible for a cyclic machine to transfer heat from a colder body (T_c) to a hotter body (T_h) without external work input (W_{in}). • **Entropy Statement:** For any spontaneous process, $\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr} > 0$ ($= 0$ at reversible equilibrium; Clausius inequality: $dS \geq đq/T$).
2. General Ideal Gas Entropy Formulas	• **Variable (T, V):** $\Delta S = n\,C_v \ln(T_f / T_i) + n\,R \ln(V_f / V_i)$ • **Variable (T, P):** $\Delta S = n\,C_p \ln(T_f / T_i) + n\,R \ln(P_i / P_f)$ • **Phase Transition at Constant T_m or T_b:** $\Delta S_{fus} = \Delta H_{fus} / T_m$ and $\Delta S_{vap} = \Delta H_{vap} / T_b$.

3. The 4-Stroke Carnot Cycle & Efficiency (10-Mark Derivation)	1. **Stroke AB (Isothermal Expansion at Hot Source T_h):** $q_h = -w_1 = +nRT_h \ln(V_2/V_1)$; $\Delta S_1 = +q_h/T_h$. 2. **Stroke BC (Adiabatic Expansion $T_h \rightarrow T_c$):** $q_2 = 0, \Delta S_2 = 0$; $w_2 = nC_v(T_c - T_h)$; $(T_h/T_c) = (V_3/V_2)^{\gamma-1}$. 3. **Stroke CD (Isothermal Compression at Cold Sink T_c):** $q_c = -w_3 = -nRT_c \ln(V_3/V_4) < 0$; $\Delta S_3 = q_c/T_c$. 4. **Stroke DA (Adiabatic Compression $T_c \rightarrow T_h$):** $q_4 = 0, \Delta S_4 = 0$; $w_4 = nC_v(T_h - T_c) = -w_2$; $(T_h/T_c) = (V_1/V_4)^{\gamma-1}$. • **Efficiency & COP:** Since $V_2/V_1 = V_3/V_4$ and cyclic entropy $\oint dS = q_h/T_h + q_c/T_c = 0$: $\eta = \lvert W_{net}\rvert / q_h = (q_h - \lvert q_c\rvert) / q_h = 1 - (T_c / T_h)$ | Refrigerator $COP_{ref} = T_c / (T_h - T_c)$.

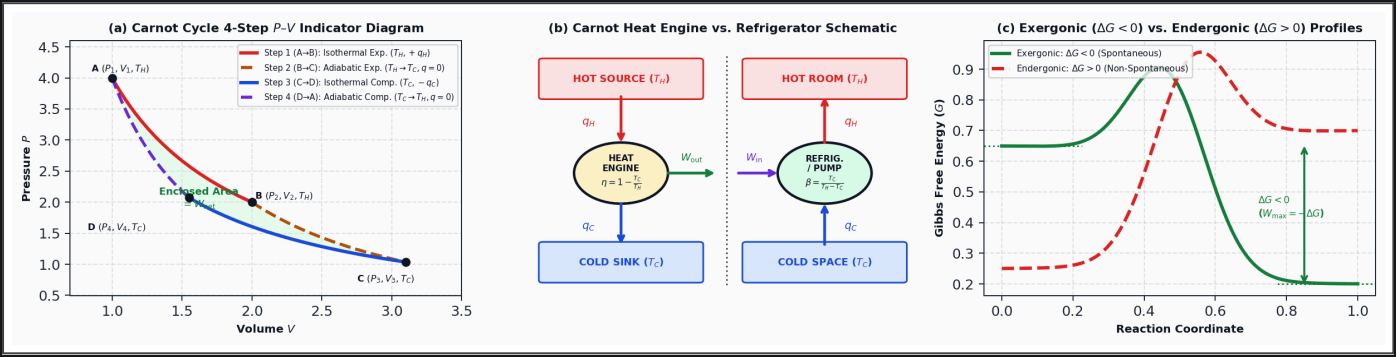


Figure 1.3: (a) 4-stroke Carnot cycle on a P–V diagram; (b) Heat engine vs. reverse Carnot refrigerator; (c) Exergonic (ΔG < 0) vs. endergonic (ΔG > 0) free energy curves.

M1S4 Free Energy (ΔA, ΔG), Spontaneity Table & Third Law (4 spontaneity regimes)

Helmholtz Work Function: $A = U - TS$ ($-\Delta A_{T,V} = w_{max,total}$; spontaneous at constant T, V when **$\Delta A < 0$**).

Gibbs Free Energy & Gibbs–Helmholtz Equation: $G = H - TS$ ($-\Delta G_{T,P} = w_{max,useful}$; spontaneous at constant T, P when **$\Delta G < 0$**): $\Delta G = \Delta H - T\Delta S = -T\Delta S_{universe}$ \| At equilibrium ($\Delta G = 0$): $\Delta G° = -RT \ln K_{eq} = -2.303\ RT \log_{10} K_{eq}$ \| Crossover Temp: $T_{eq} = \Delta H / \Delta S$.

Third Law of Thermodynamics: The entropy of a pure, perfectly crystalline solid approaches zero at absolute zero ($T = 0$ K): $S(0\ K) = k_B \ln(1) = 0$.

ΔH° Sign	ΔS° Sign	ΔG° = ΔH° − TΔS°	Spontaneity Condition	Slide & Exam Examples
− (Exothermic)	**+ (Disorder ↑)**	Always **− (< 0)**	Spontaneous at **ALL temperatures**	Combustion; $2O_3(g) \rightarrow 3O_2(g)$
− (Exothermic)	**− (Order ↑)**	**−** at low T; **+** at high T	Spontaneous at **Low T (T < ΔH/ΔS)**	Freezing of water; Haber $N_2+3H_2 \rightarrow 2NH_3$
+ (Endothermic)	**+ (Disorder ↑)**	**+** at low T; **−** at high T	Spontaneous at **High T (T > ΔH/ΔS)**	Melting of ice (>0°C); $CaCO_3 \rightarrow CaO+CO_2$
+ (Endothermic)	**− (Order ↑)**	Always **+ (> 0)**	**Non-spontaneous** at all temperatures	$3O_2(g) \rightarrow 2O_3(g)$; $N_2+2O_2 \rightarrow 2NO_2$

M1S5 Chemical Kinetics: Rate Laws, Order vs. Molecularity & Integrated Laws (4 orders)

For $aA + bB \rightarrow cC + dD$: **Rate** $= -(1/a)d[A]/dt = -(1/b)d[B]/dt = +(1/c)d[C]/dt = +(1/d)d[D]/dt = k[A]^m[B]^n$.

• **Order (m + n):** Sum of concentration exponents in the experimental rate law (can be **0, integer, or fractional**; applies to overall complex reactions).

• **Molecularity:** Number of reacting species (1 = unimolecular, 2 = bimolecular, 3 = termolecular) colliding simultaneously in a single **elementary step** (always a **positive integer ≤ 3**; never 0 or fractional).

• **General Half-Life Scaling Rule:** $t_{1/2} \propto [A]_0^{1-n} \Rightarrow (t_{1/2})_1 / (t_{1/2})_2 = ([A]_{0,2} / [A]_{0,1})^{n-1}$.

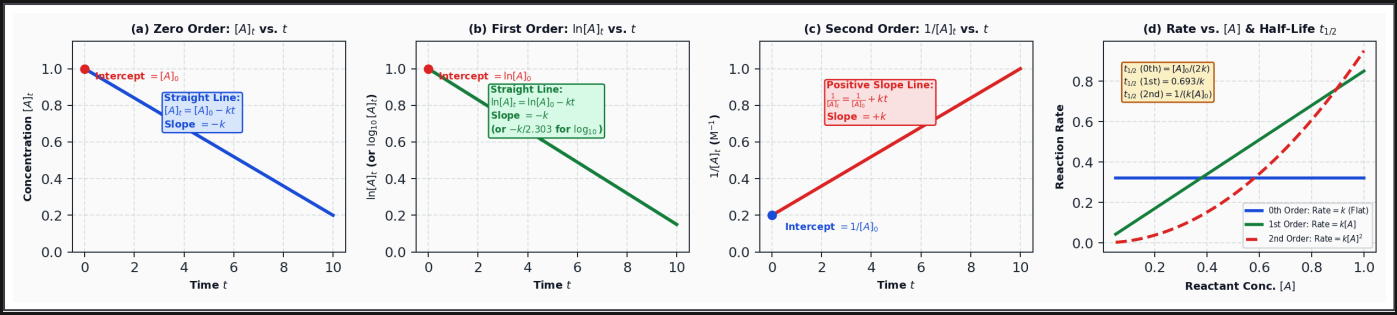


Figure 1.4: (a) Linear integrated rate plots ($[A]_t$, $\ln[A]_t$, and $1/[A]_t$ vs. t); (b) Rate vs. concentration curves; (c) Half-life $t_{1/2}$ scaling for Zero, First, and Second order reactions.

Reaction Order	Differential Law	Integrated Rate Law	Linear Plot & Slope	Half-Life ($t_{1/2}$)	Units of k & Examples
Zero Order ($n = 0$)	**$-d[A]/dt = k$**	**$[A]_t = [A]_0 - kt$** $k = ([A]_0 - [A]_t) / t$	$[A]_t$ vs. t Slope = **$-k$**	**$t_{1/2} = [A]_0 / (2k)$** ($t_{1/2} \propto [A]_0$)	**$mol\ L^{-1}\ s^{-1}$** NH_3 on hot Pt/W; photochemical H_2+Cl_2
First Order ($n = 1$)	**$-d[A]/dt = k[A]$**	**$\ln[A]_t = \ln[A]_0 - kt$** $k = (2.303/t)\ \log([A]_0/[A]_t)$	$\ln[A]_t$ vs. t Slope = **$-k$**	**$t_{1/2} = 0.693 / k$** (Indep. of $[A]_0$!)	**s^{-1} (or min^{-1})** $CH_3NC \rightarrow CH_3CN$; C_4H_9Cl; N_2O_5
Pseudo-1st Order	**$-d[A]/dt = k_{obs}[A]$** ($k_{obs} = k'[B]_0$)	**$k_{obs} = (2.303/t)\ \log([A]_0/[A]_t)$** True $k' = k_{obs} / [B]_0$	$\ln[A]_t$ vs. t Slope = **$-k_{obs}$**	**$t_{1/2} = 0.693 / k_{obs}$**	Acid hydrolysis of ester $CH_3COOC_2H_5$ or CH_3I in excess H_2O
Second Order ($n = 2$)	**$-d[A]/dt = k[A]^2$**	**$1/[A]_t = 1/[A]_0 + kt$** $k = (1/t)(1/[A]_t - 1/[A]_0)$	$1/[A]_t$ vs. t Slope = **$+k$**	**$t_{1/2} = 1 / (k[A]_0)$** ($t_{1/2} \propto 1/[A]_0$)	**$L\ mol^{-1}\ s^{-1}$ ($M^{-1}s^{-1}$)** $2NO_2 \rightarrow 2NO + O_2$; $2HI \rightarrow H_2 + I_2$

M1S6 Collision Theory, Arrhenius Equation & Catalysis (3 catalyst classes)

1. Collision Theory & Orientation Factor	• **Two Mandatory Conditions for Effective Collision:** Reacting molecules must collide with (1) kinetic energy ≥ **Activation Energy (E_a)**, and (2) proper **steric orientation (p)**: **$k = p\ Z_{AB}\ e^{-E_a/(RT)}$**. • **Slide Case Study (Cl + NOCl → NO + Cl_2):** Only when attacking **Cl•** strikes the chlorine atom of **NOCl** does **Cl_2** form; hitting the oxygen end bounces off ineffectively. • **Ionic Charge Effect (Slide 22 PYQ):** Reactant ions of **like charge** repel (raising E_a → **slower rate**); ions of **unlike (opposite) charge** attract (lowering E_a → **much faster rate**).
2. Maxwell–Boltzmann & Arrhenius Equation	• Raising temperature from **T_1** to **T_2** shifts the Maxwell–Boltzmann distribution tail to higher energies, exponentially increasing the fraction of molecules **$f = e^{-E_a/(RT)}$** above **E_a**: **$k = A\ e^{-E_a/(RT)}$** ⇒ **$\ln k = \ln A - (E_a / R)(1 / T)$** ⇒ **$\ln(k_2 / k_1) = (E_a / R)\ [\ (T_2 - T_1) / (T_1\ T_2)\ ]$**.

3. Three Classes of Catalysis (Lowers E_a; ΔH, ΔG, K_{eq} unchanged!)	1. **Homogeneous Catalysis (Same Phase):** $Cl_2(g)$ catalyzing $2N_2O(g) \rightarrow 2N_2(g) + O_2(g)$ via $Cl^\bullet$ / $ClO^\bullet$; mineral acid $H^+(aq)$ catalyzing ester/CH_3I hydrolysis. 2. **Heterogeneous Catalysis (Different Phase — Adsorption → Reaction → Desorption):** Solid **Fe(s)** in Haber synthesis ($N_2 + 3H_2 \rightarrow 2NH_3$); solid V_2O_5 / **Pt** in Contact process ($2SO_2 + O_2 \rightarrow 2SO_3$); $MnO_2(s)$ decomposing $2H_2O_2(aq)$. 3. **Enzyme (Biological) Catalysis (Michaelis–Menten Lock-and-Key: E + S ↔ ES → E + P):** Liver *catalase* decomposing $2H_2O_2 \rightarrow 2H_2O + O_2$ lowers E_a from **75 kJ/mol** to **~8 kJ/mol**.

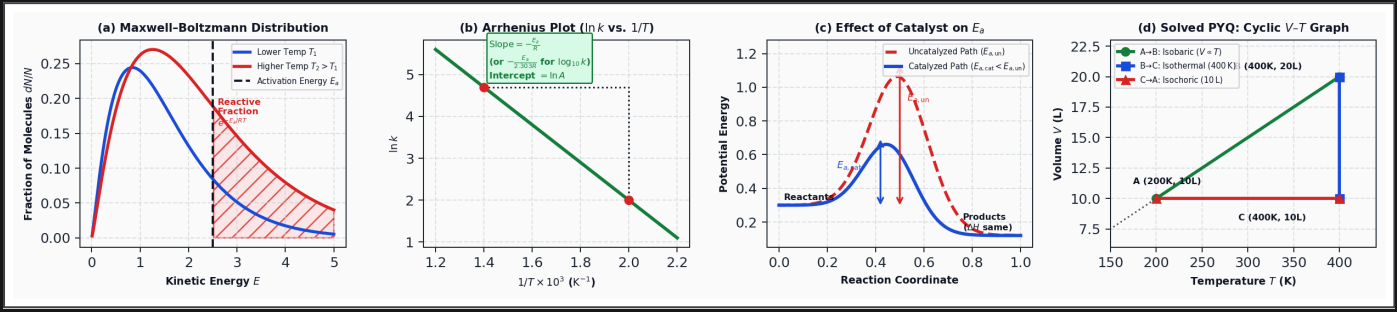


Figure 1.5: (a) Maxwell–Boltzmann distribution at T_1 vs. T_2; (b) Linear Arrhenius plot (Slope = $-E_a/R$); (c) Uncatalyzed vs. catalyzed energy barrier; (d) CAT-1 cyclic V–T triangle A → B → C.

M1S7 Complete Solved Numericals & CAT-1 / FAT PYQ Master Bank (10 solved problems)

PYQ 1: Steam Vaporization Work & ΔU (Slide 2 & Slide 24)	**Prompt:** Calculate *w* and *ΔU* when 1 mol of liquid water at 100 °C (373 K) vaporizes into steam at 1 atm ($\Delta H_{vap} = 40{,}670$ J/mol). **Solution:** Since $V_{steam} \gg V_{liquid}$, $w = -P\Delta V \approx -nRT = -(1)(8.314)(373) = -3101.1$ J. At const P, $q_P = \Delta H_{vap} = +40{,}670$ J. By 1st Law: $\Delta U = q + w = 40{,}670 - 3101.1 = +37{,}568.9$ J (+37.57 kJ/mol).
PYQ 2: CAT-1 Cyclic V–T Triangle (Slides 14–16 Q4 & Q7 — 10 Marks)	**Prompt:** 1 mol monoatomic ideal gas ($C_v = 1.5R$, $C_p = 2.5R$) traverses cycle **A(200K, 10L) → B(400K, 20L, V∝T) → C(400K, 10L) → A(200K, 10L)** (Figure 1.5d). Find *q, w, ΔU, ΔH, ΔS* for each step and total cycle. **Solution:** • **Step A → B (Isobaric, since V/T = const):** $w_{AB} = -nR(T_B - T_A) = -8.314(200) = -1662.8$ J; $\Delta U_{AB} = nC_v\Delta T = +2494.2$ J; $q_{AB} = \Delta H_{AB} = nC_p\Delta T = +4157.0$ J; $\Delta S_{AB} = nC_p \ln(400/200) = +14.41$ J/K. • **Step B → C (Isothermal at 400 K, 20L → 10L):** $\Delta U_{BC} = \Delta H_{BC} = 0$; $w_{BC} = -nRT_B \ln(10/20) = +2305.1$ J; $q_{BC} = -2305.1$ J; $\Delta S_{BC} = R \ln(0.5) = -5.76$ J/K. • **Step C → A (Isochoric at 10 L, 400K → 200K):** $w_{CA} = 0$; $q_{CA} = \Delta U_{CA} = nC_v(-200) = -2494.2$ J; $\Delta H_{CA} = -4157.0$ J; $\Delta S_{CA} = C_v \ln(0.5) = -8.65$ J/K. • **Cycle Totals:** $\Delta U_{cyc} = \Delta H_{cyc} = \Delta S_{cyc} = 0$; $w_{cyc} = +642.3$ J; $q_{cyc} = -642.3$ J. *(For the 300K → 600K, 12L → 24L Q7 variant: multiply all energies by 1.5 ⇒ $w_{AB} = -2494.2$ J, $w_{BC} = +3457.7$ J, $w_{CA} = 0$, $w_{cyc} = +963.5$ J).*
PYQ 3: Isothermal Final Volume & Irrev. Work (Slides 19–22 Q1)	• **(a)** 1 mol ideal gas at STP (273.15 K, $V_i = 22.4$ L) absorbs $q_{rev} = 1000$ cal in a reversible isothermal expansion: $\ln(V_f/22.4) = q_{rev}/(nRT) = 1000/(1 \times 1.987 \times 273.15) = 1.8425 \Rightarrow V_f = 22.4 \times e^{1.8425} = 141.4$ L. • **(b)** 10 mol gas expands isothermally against $P_{ex} = 1$ atm from 15 L to 35 L: $w = -P_{ex}\Delta V = -1(20)$ L·atm × 101.325 J/(L·atm) = −2026.5 J. • **(c)** Gas expands against 1 atm from 30 L to 60 L absorbing $q = +700$ J: $w = -30 \times 101.325 = -3039.8$ J ⇒ $\Delta U = 700 - 3039.8 = -2339.8$ J. • **(d)** 1 mol N_2 at 300 K expands against $P_{ex} = 5$ atm from 1 L to 3 L: $\Delta U = \Delta H = 0$; $w = -5(2) \times 101.325 = -1013.25$ J; $q = +1013.25$ J.

PYQ 4: Spontaneity (Ice, CO & $CaCO_3$) (Slides 7, 18, 21)	• **(a) Melting of Ice at −10 °C (263 K):** Given **$\Delta H = +6.01$ kJ/mol, $\Delta S = +0.022$ kJ/K·mol**: **$\Delta G = 6.01 - 263(0.022) = +0.224$ kJ/mol > 0** (**Non-spontaneous** at −10 °C; freezing is spontaneous!). • **(b) $2CO + O_2 \rightarrow 2CO_2$ at 298.15 K:** Given **$\Delta H = -128.3$ kJ, $\Delta S = -159.5$ J/K**: **$\Delta G = -128{,}300 - 298.15(-159.5) = -80.75$ kJ < 0** (Spontaneous below **$T_{eq} = \Delta H/\Delta S = 804.4$ K**). • **(c) $CaCO_3 \rightarrow CaO + CO_2$:** Given **$\Delta H° = +178.3$ kJ/mol, $\Delta S° = +160.5$ J/K·mol**: Spontaneous above **$T_{eq} = 178{,}300 / 160.5 = 1110.9$ K (837.8 °C)**.
PYQ 5: Kinetics & Arrhenius Calculations (Slides 10, 22, 23 & FAT)	• **(a) C_4H_9Cl 1st-Order Decay:** Drops from **0.100 M to 0.050 M** in **340 s (= $t_{1/2}$)** ⇒ **$k = 0.693 / 340 = 2.04 \times 10^{-3}$ s^{-1}**. • **(b) Bimolecular Rate (Slide 22 & 23):** **Rate = k[A][B]**. For **$k = 5.7\times10^{-2}$, [A] = 5.0×10^{-3}, [B] = 2.0×10^{-2}** ⇒ **Rate = 5.7×10^{-6} M s^{-1}**. For **$k = 4.2\times10^{-3}$, [A] = 5.5×10^{-2}, [B] = 2.9×10^{-2}** ⇒ **Rate = 6.70×10^{-6} M s^{-1}**. • **(c) Arrhenius Rate Constant at 300 K (Slide 22):** Given **$E_a = 30$ kJ/mol, $A = 5.0\times10^{10}$**: **$k = 5.0\times10^{10} \exp(-30{,}000 / [8.314 \times 300]) = 2.99 \times 10^5$ M^{-1} s^{-1}**. • **(d) Two-Temp Arrhenius (FAT):** Rate quadruples (**$k_2/k_1 = 4$**) from **300 K to 320 K**: **$E_a = 8.314 \times \ln(4) \times (300 \times 320)/20 = 55.32$ kJ/mol**.